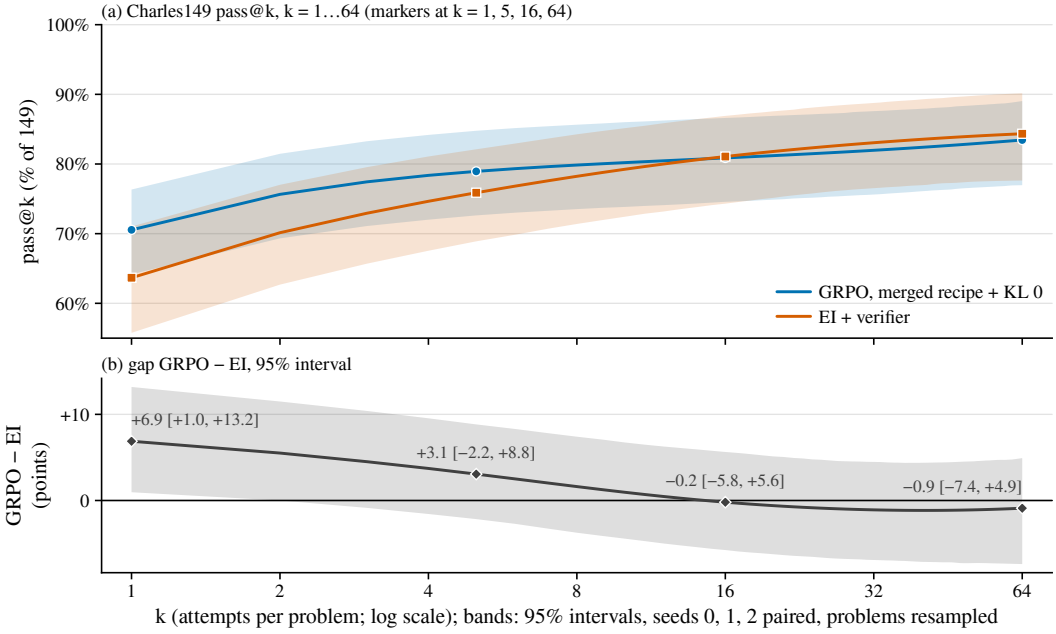
(a) Charles149 pass@k, k = 1…64 (markers at k = 1, 5, 16, 64)
100%
90%
80%
70%
60%
pass@k (% of 149)
GRPO, merged recipe + KL 0
EI + verifier
(b) gap GRPO − EI, 95% interval
+10
0
GRPO − EI (points)
+6.9 [+1.0, +13.2]
+3.1 [−2.2, +8.8]
−0.2 [−5.8, +5.6]
−0.9 [−7.4, +4.9]
1
2
4
8
16
32
64
k (attempts per problem; log scale); bands: 95% intervals, seeds 0, 1, 2 paired, problems resampled